

Marketing Analytics

Chapter 4 · Learning from Experiments

Nils Wlömert

Vienna University of Economics and Business (WU)

September 2026

Part 1 · Experiments I

Today, part B

Experiments I (session 3)

- 4.1 Back to the A/B test
- 4.2 The difference between two means
- 4.3 Testing a hypothesis
- 4.4 How large is the effect?
- 4.5 Reporting the result

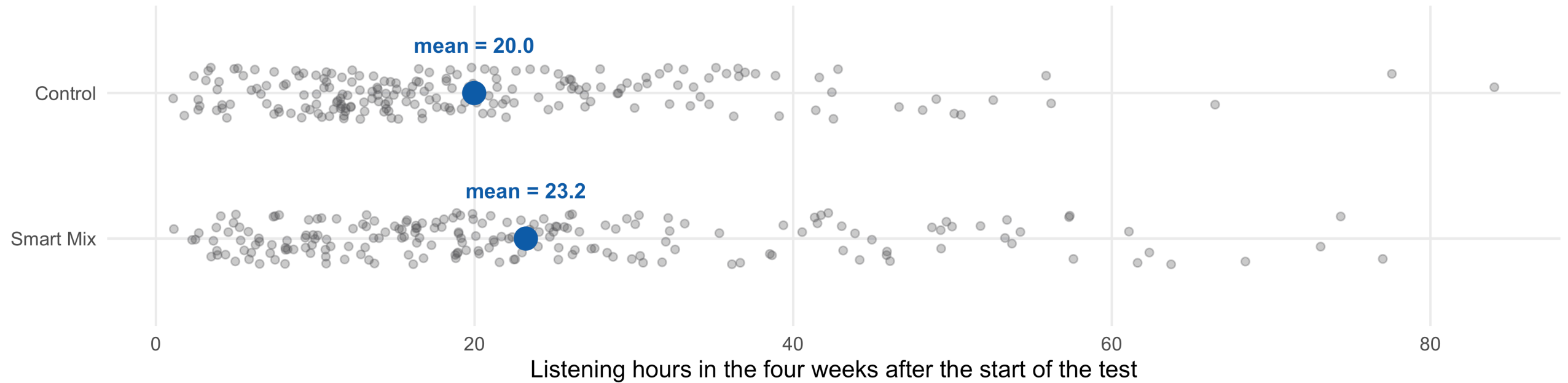
Experiments II (session 4)

- 4.6 Comparing shares
- 4.7 Errors, power, and sample size
- 4.8 Other designs and tests
- 4.9 Using p -values responsibly

Driving question: *Did it work?*

4.1 Back to the A/B test

Smart Mix: +3.2 hours



```
1 t.test(hours ~ group, data = ab_test)
```

Welch Two Sample t-test

data: hours by group

t = 2.1617, df = 386.96, p-value = 0.03125

alternative hypothesis: true difference in means between group Smart Mix and group Control is not equal to 0

95 percent confidence interval:

0.293571 6.196096

sample estimates:

mean in group Smart Mix	mean in group Control
23.21980	19.97497

Three questions for the product team

1. **Could the difference be chance?** Two random groups never have exactly the same mean — is 3.2 hours more than chance typically produces?
2. **How large is the effect — and does it matter?**
3. **Was the experiment large enough?** How large should the next one be?

The formulas stay minimal: we compute **one** standard error by hand to see what a test does. After that, R does the arithmetic — we do the interpretation.

4.2 The difference between two means

Signal and noise

Repeat the experiment many times → the **difference** varies too. Its sampling distribution is:

1. **Centered** on the true effect
2. **Approximately normal** (central limit theorem)
3. As wide as the **standard error of the difference**:

$$SE_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Smart Mix:

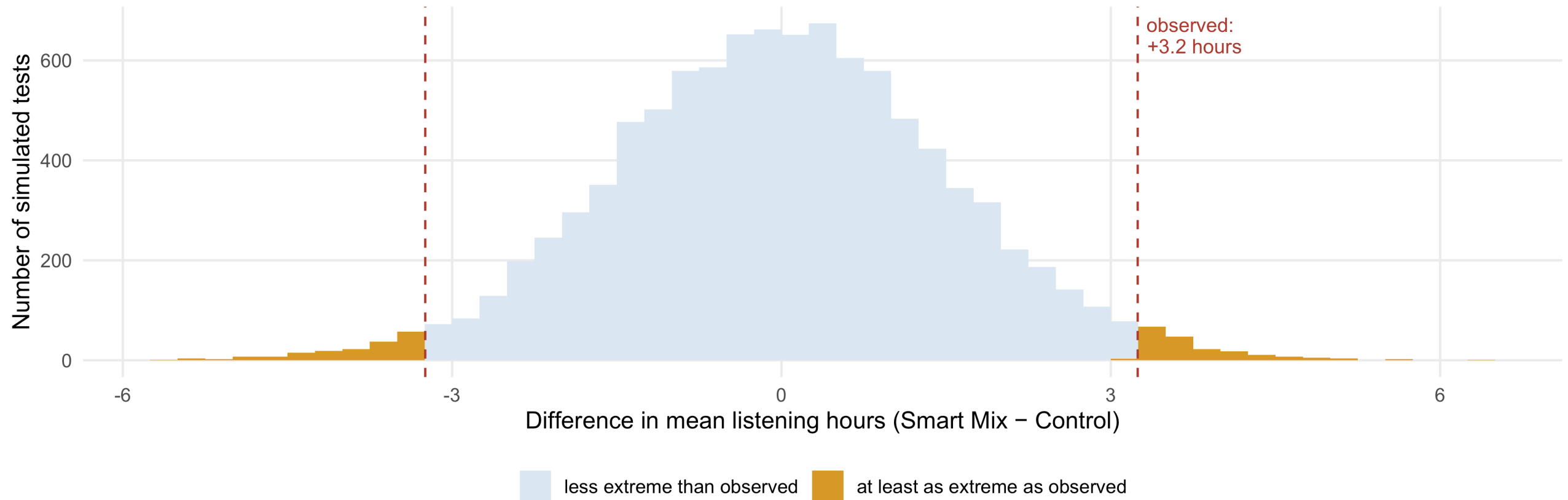
$$\sqrt{\frac{16.2^2}{200} + \frac{13.7^2}{200}} = 1.50 \text{ hours}$$

Every test compares the **signal** (observed difference) with the **noise** (the standard error). This is the only SE we compute by hand today.

A world in which Smart Mix does nothing

10,000 A/B tests in a world in which Smart Mix has no effect

3.6% of the simulated differences are at least as extreme as the observed one



If Smart Mix had **no effect**, a difference of 3.2 hours or more would occur in only ~4% of experiments. Either Smart Mix works — or the team saw a rare accident.

Live demo: how surprising is the difference?

Course book, §4.2 — change the **users per group** and the **observed difference**.

<https://wu-rds.github.io/MA2026/04-experiments.html#sec-exp-difference>

Screenshot placeholder: the “null world” widget
`img/ch04/widget_null_world.png`

Is a difference of 3.2 hours surprising with 50 users per group? With 800?

4.3 Testing a hypothesis

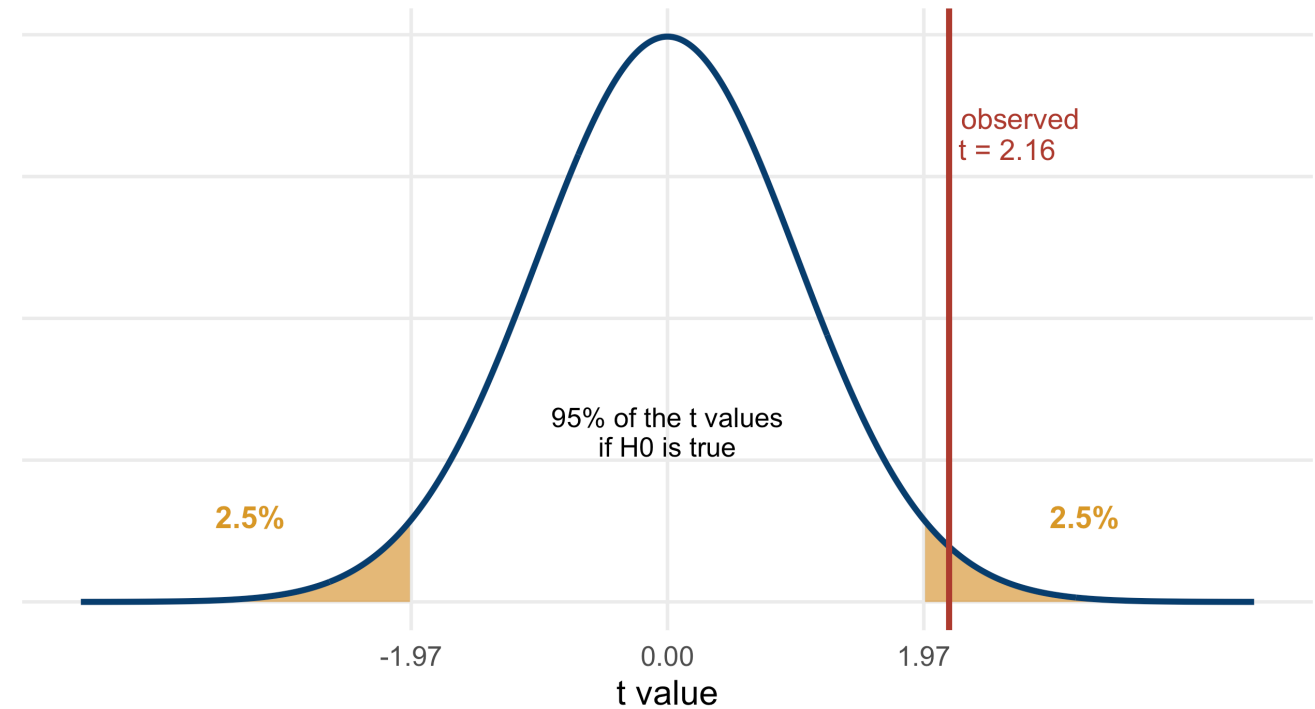
Null hypothesis and test statistic

- H_0 : no effect, $\mu_1 = \mu_2$
- H_1 : an effect, $\mu_1 \neq \mu_2$

Test statistic = signal / noise:

$$t = \frac{3.24 - 0}{1.50} = 2.16$$

Where does the observed t value fall if Smart Mix has no effect?



If H_0 were true, 95% of all t values would lie between -1.97 and $+1.97$. Ours lies outside.

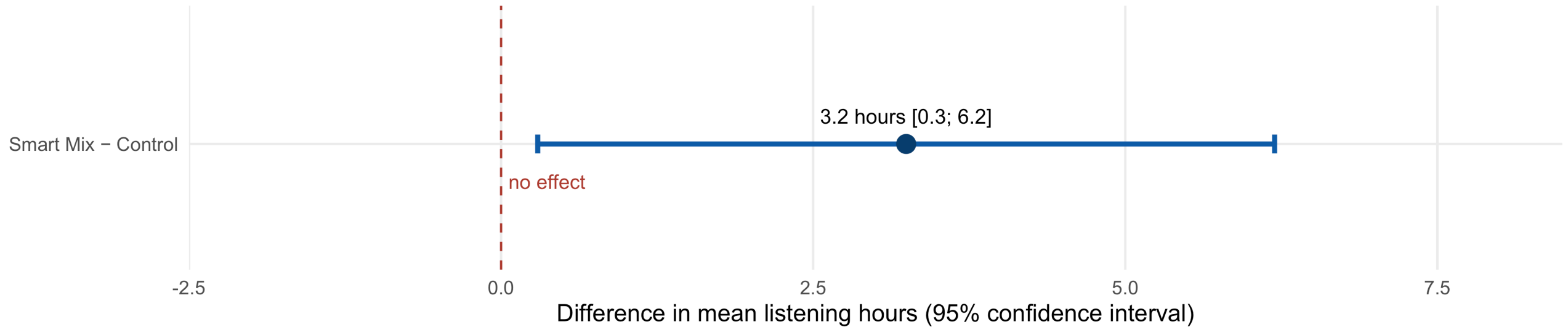
The p -value and the decision

p-value: the probability of a difference at least as extreme as the observed one — if H_0 were true.

- Smart Mix: $p = 0.031$ — in a no-effect world, ~3 in 100 experiments would show ≥ 3.2 hours
- Decision rule with significance level $\alpha = 5\%$ (chosen **in advance**):
 - $p < \alpha \rightarrow$ **statistically significant**, reject H_0
 - $p \geq \alpha \rightarrow$ **do not reject H_0** — *not* “there is no effect”

Our simulation said ~4%, the t -test says 3.1% — the test formalizes the thought experiment.

The confidence interval of the difference



CI excludes 0 $\Leftrightarrow p < 0.05$. Test and interval always agree.

The interval tells more: the effect could be a quarter of an hour — or six hours.

Reading the output

Output	Meaning
Welch Two Sample t-test	Two independent groups; equal variances not assumed (R's safe default)
$t = 2.1617$	Observed difference = 2.16 standard errors away from zero
$df = 386.96$	Degrees of freedom — shape of the t -distribution; irrelevant for large samples
$p\text{-value} = 0.03125$	With no effect, $\geq$ this difference in 3.1% of experiments
alternative hypothesis: ...	Two-sided: a difference in either direction
95 percent confidence interval: 0.29 6.20	Effects compatible with the data
mean in group ...	Group means; their difference = the estimated effect

Your turn (5 min)

Run the test again with only the **first 50 users per group**:

```
1 t.test(hours ~ group,  
2       data = ab_test |> group_by(group) |> slice_head(n = 50))
```

How do the **difference**, the **confidence interval**, and the ***p*-value** change? Explain with the standard error.

4.4 How large is the effect?

Significant $\neq$ large $\neq$ important

- **Original unit:** +3.2 hours in four weeks
- **Relative:** +16% vs. control (20.0 hours)
- **Standardized:** Cohen's d = difference / pooled SD

```
1 group_stats <- ab_test |> group_by(group) |> summarise(mean = mean(hours  
2 diff(rev(group_stats$mean)) / sqrt(mean(group_stats$sd^2)))
```

```
[1] 0.2161689
```

Conventions (Cohen, 1992): 0.2 small · 0.5 medium · 0.8 large

$d = 0.22$: statistically **small** — but +16% listening time can be worth a lot to a service with millions of users. The conventions are a starting point, not a verdict.

4.5 Reporting the result

Effect first, uncertainty second, test last

Users with Smart Mix listened on average 23.2 hours in four weeks, compared with 20.0 hours in the control group — an increase of 3.2 hours (+16%, 95% CI [0.3; 6.2] hours). The difference is statistically significant (Welch t -test: $t(387) = 2.16$, $p = .031$) but small in standardized terms ($d = 0.22$). The confidence interval is wide: the true effect could be anywhere between a few minutes and six hours.

The last sentence is the most important one: it tells the reader **how much the conclusion is worth**.

Part 2 · Experiments II

Before we start

Debrief Assignment 1 *(10 min)*

- Three short paper questions about *your* results
- Closed book, no laptop

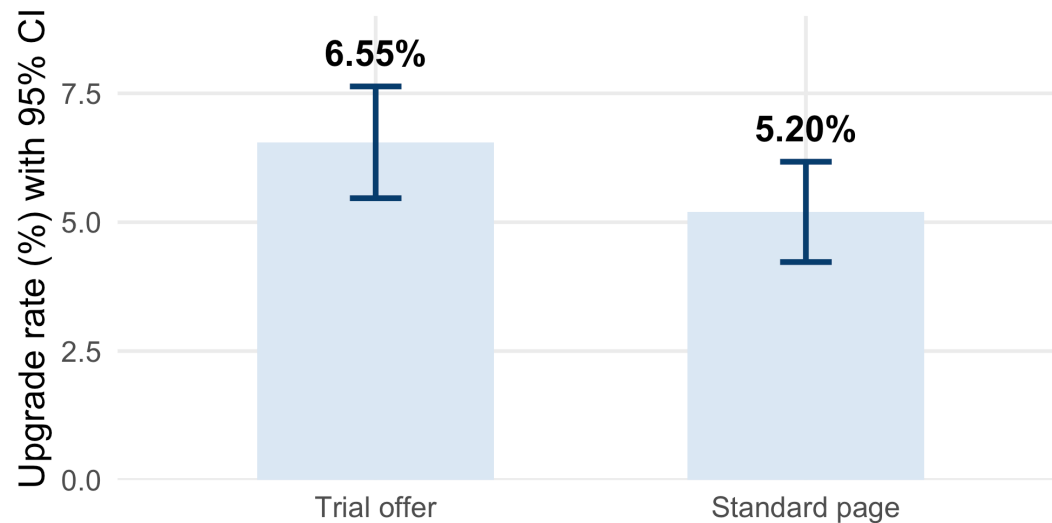
Paper quiz on Chapter 3 *(10 min)*

- Individual, closed book
- Counts toward participation

4.6 Comparing shares

A Premium trial offer

- New Premium page with a **one-month free trial** vs. standard page
- 2,000 free users per group, randomized
- Outcome: **upgrade within two weeks** (yes/no)



```
1 prop.test(x = c(131, 104), n = c(2000, 2000))
```

2-sample test for equality of proportions with continuity correction

```
data:  c(131, 104) out of c(2000, 2000)
X-squared = 3.0561, df = 1, p-value = 0.08043
alternative hypothesis: two.sided
95 percent confidence interval:
 -0.001568859  0.028568859
sample estimates:
prop 1 prop 2
0.0655 0.0520
```

“Not significant” — so it does not work?

- +1.35 percentage points (+26% relative) — but $p = 0.08$
- 95% CI of the difference: **-0.16 to +2.86** points
- Compatible with no effect — **and** with a substantial one

The right conclusion: “**this experiment was too small to tell**” — not “the trial offer does not work”.
(`prop.test()` = chi-square test for a 2×2 table; `chisq.test()` for more groups or categories.)

Your turn (5 min)

Two sales agents each talk to **300 customers**. Jon gets **200** to buy, Mark **100**.

1. Is Jon's conversion rate significantly higher? Use `prop.test()` and report the difference with its 95% CI.
2. Does this show that Jon is the **better salesperson**?

Example after McDonnell Feit & McCullough, *Business Experiments*.

4.7 Errors, power, and sample size

Two ways to be wrong

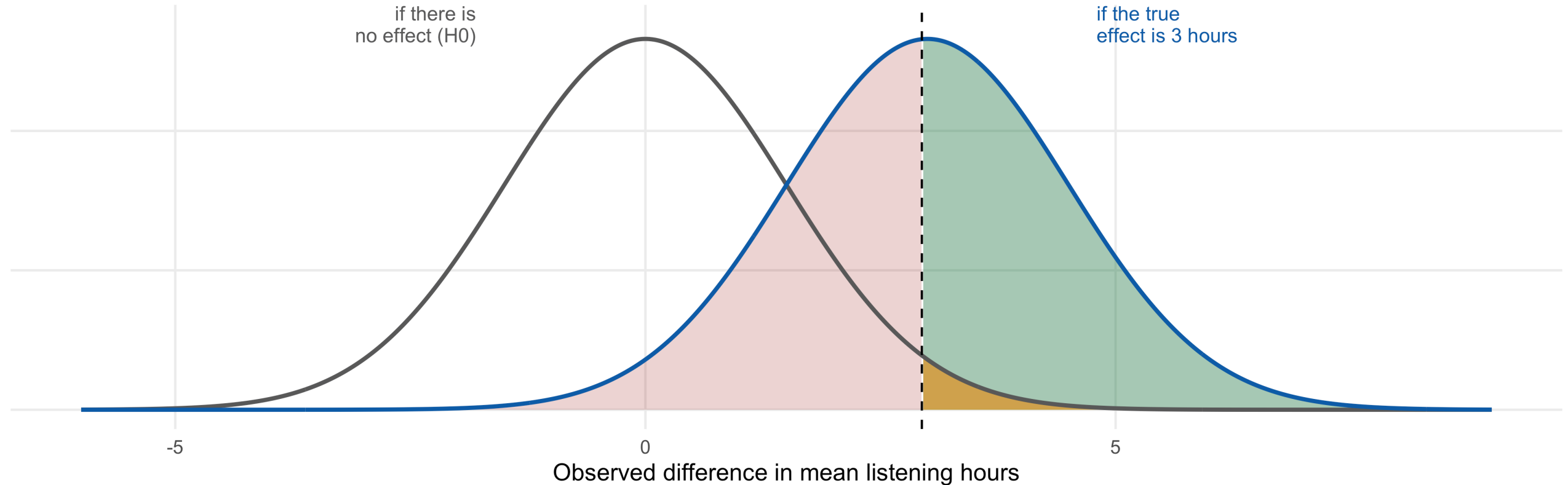
	No effect in reality (H_0 true)	Effect in reality (H_0 false)
Significant	Type I error — false alarm (α)	✓ effect detected (power = $1 - \beta$)
Not significant	✓	Type II error — effect missed (β)

- $\alpha = 5\%$: one in twenty tests of a useless feature is “significant” anyway
- Power = probability of detecting a real effect; common target: **80%**
- Lower $\alpha \rightarrow$ lower power. Fewer errors of **both** kinds $\rightarrow$ only with **larger samples**

What determines power

Power = 52% (n = 200 per group)

Green: effect detected (power) · red: effect missed (type II error) · orange: false alarm (type I error)



Effect size · sample size · variability. With a true effect of 3 hours, Smart Mix had only ~55% **power** — the team was somewhat lucky.

Live demo: what determines power?

Course book, §4.7 — change the **true effect**, the **users per group**, and **α** .

<https://wu-rds.github.io/MA2026/04-experiments.html#sec-exp-power>

Screenshot placeholder: the power widget
`img/ch04/widget_power.png`

Halve the true effect. How many more users do you need for the same power?

Sample size: decide before you start

Means — Smart Mix, $d = 0.22$, 80% power:

```
1 power.t.test(delta = 0.22, sd = 1, power = 0.80)$n
```

```
[1] 325.297
```

Shares — Premium test, 5% → 6%:

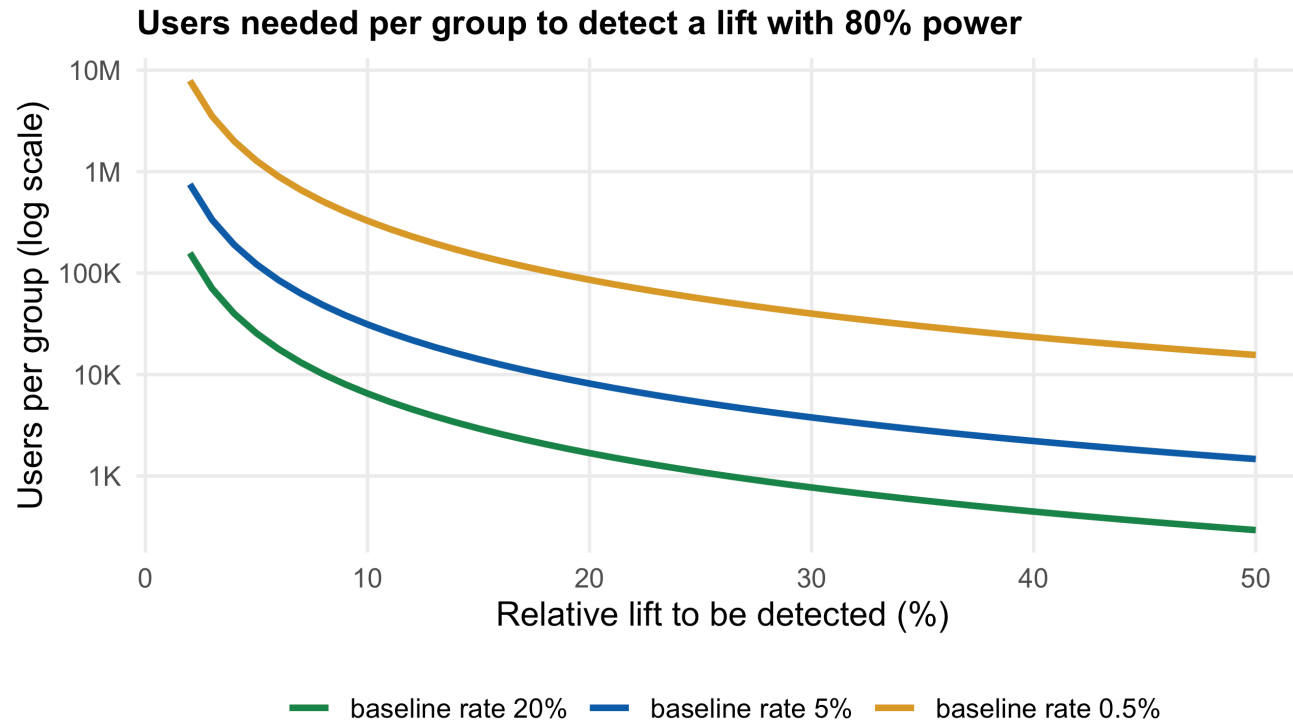
```
1 power.prop.test(p1 = 0.05, p2 = 0.06, power = 0.80)$n
```

```
[1] 8157.731
```

Start from the **smallest effect that would still matter** for the decision — not from the effect you hope for.

- Smart Mix: ~330 per group (not 200)
- Premium test: > 8,000 per group (not 2,000)
- Omit **power** and give **n** instead → the power of the test you ran

Why online advertising needs huge experiments



- CTR 0.50% $\rightarrow$ 0.55% (+10%): > 300,000 users per group
- Lewis & Rao (2015): even experiments with **millions** of users are often too imprecise to tell whether ads pay off
- Remember eBay (\$1.2): the whole US market

Low baseline rates + small lifts = **enormous samples.**

Lewis & Rao (2015), "The Unfavorable Economics of Measuring the Returns to Advertising", *Quarterly Journal of Economics*, 130(4), 1941–1973.

Live demo: how many users does a conversion test need?

Course book, §4.7 — baseline rate, lift, and traffic per day.

<https://wu-rds.github.io/MA2026/04-experiments.html#sec-exp-power>

Screenshot placeholder: the sample-size widget
`img/ch04/widget_sample_size.png`

Your group project: ~60 per condition → only $d \geq 0.5$ detectable with 80% power (small effects: < 20% power). **Make your manipulation strong.**

Your turn (5 min)

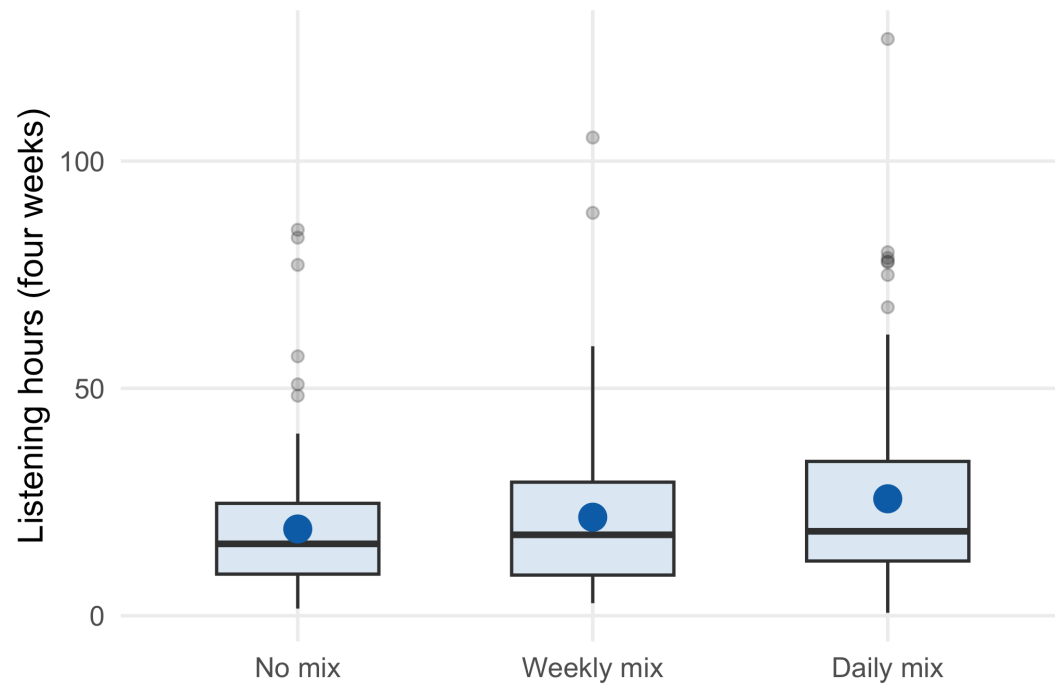
A newsletter has a click-through rate of **2%**. A new subject line should be tested.

1. How many recipients per version for detecting **+10%** (2.0% → 2.2%) with 80% power?
2. And if only **+25%** matters?

Use `power.prop.test()`.

4.8 Other designs and tests

More than two groups: ANOVA



```
1 summary(aov(hours ~ version, data = mix_test))
```

```
version      Df Sum Sq Mean Sq F value    Pr(>F)
Residuals  447 131755    294.8
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
1 pairwise.t.test(mix_test$hours, mix_test$version,
2                 p.adjust.method = "bonferroni")
```

Pairwise comparisons using t tests with pooled SD

data: mix_test\$hours and mix_test\$version

```
           No mix Weekly mix
Weekly mix 0.5732 -
Daily mix  0.0026 0.1247
```

P value adjustment method: bonferroni

Three t -tests? $1 - 0.95^3 = 14\%$ chance of at least one false alarm (10 tests: 40%)

Same people, both conditions: paired test

```
1 search_test <- search_data() # 60 users, old vs. new search design
2 t.test(search_test$new_design, search_test$old_design, paired = TRUE)$p.value
```

```
[1] 0.0002713375
```

```
1 t.test(search_test$new_design, search_test$old_design)$p.value # wrong test
```

```
[1] 0.1392638
```

Each user is **their own control**: the large differences *between* users drop out. The new design saves ~2 seconds — invisible in the wrong (independent) test.

Choosing a test

Outcome	Groups	Design	Test	R
Metric	2	different people	Welch <i>t</i> -test	<code>t.test(y ~ group)</code>
Yes/no	2+	different people	proportions / chi-square	<code>prop.test(), chisq.test()</code>
Metric	2	same people	paired <i>t</i> -test	<code>t.test(x1, x2, paired = TRUE)</code>
Metric	3+	different people	ANOVA + post-hoc	<code>aov(), pairwise.t.test()</code>
Ordinal / extreme skew, small <i>n</i>	2	different people	Wilcoxon rank-sum	<code>wilcox.test(y ~ group)</code>
Ordinal / extreme skew, small <i>n</i>	3+	different people	Kruskal–Wallis	<code>kruskal.test(y ~ group)</code>

4.9 Using p -values responsibly

What a p -value is — and what it is not

$$p = P(\text{data} \mid H_0)$$

“If there were no effect, how likely would data like ours be?”

Not: $P(H_0 \mid \text{data})$

$$P(\text{runny nose} \mid \text{cold}) \neq P(\text{cold} \mid \text{runny nose})$$

$p = 0.031$. True or false?

1. 3.1% probability that Smart Mix has no effect
2. 96.9% probability that Smart Mix works
3. The effect is important

Significant $\neq$ important · not significant $\neq$ no effect

- **Significance depends on n:** with millions of users, almost everything is “significant”
- **Absence of evidence $\neq$ evidence of absence:** the Premium test was not significant, but compatible with +3 points

Everyday comparison	d	n per group (80%)
Men taller than women	1.85	6
Spicy-food fans like Indian food more	0.80	26
Men weigh more than women	0.59	46
Science fans name more planets	0.07	3,669

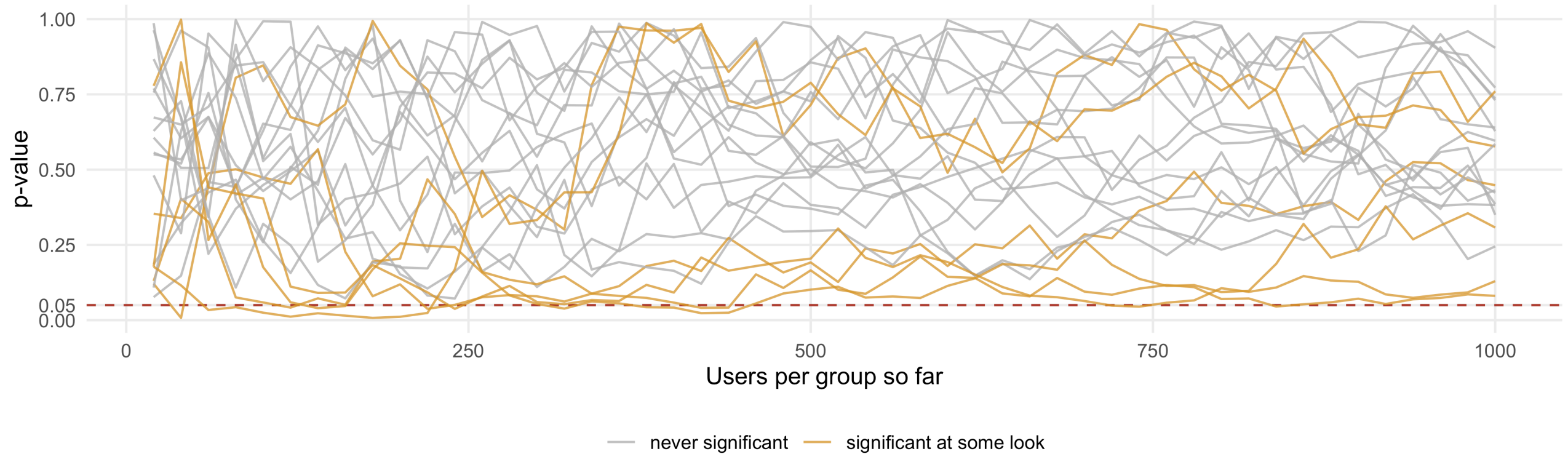
Nothing special happens at $p = 0.05$: 0.049 is no stronger evidence than 0.051. **Report the exact p , the effect size, and the CI** — and decide on all three plus costs and benefits.

Simmons, Nelson & Simonsohn (2013), survey of 697 US participants; Wasserstein & Lazar (2016), “The ASA Statement on p-Values”, *The American Statistician*.

Peeking destroys the p -value

20 A/B tests without any true effect, checked after every 20 users per group

6 of 20 tests cross $p < .05$ at some point



Checking the dashboard every day and stopping at $p < 0.05$ turns a 5% false-alarm rate into **30–40%**. **Fix the sample size in advance; evaluate once** — or use sequential testing.

Live demo: the danger of peeking

Course book, §4.9 — 200 A/B tests **without** any effect; the analyst stops at the first $p < 0.05$.

<https://wu-rds.github.io/MA2026/04-experiments.html#sec-exp-nhst>

Screenshot placeholder: the peeking widget
`img/ch04/widget_peeking.png`

What happens to the false-alarm rate if the analyst checks **more often**, or lets the test run **longer**?

Many metrics, many segments, many tests

- 20 metrics or segments, no true effect anywhere:
 $1 - 0.95^{20} = 64\%$ chance of at least one “significant” result
- Reporting only the hits = **p-hacking**
- Paying attention only to “successes” = **publication bias**

*Image placeholder: publication bias
graphic*
[img/ch04/publication_bias.png](#)

Good practice for experiments

1. **Before:** outcome, hypothesis, smallest relevant effect, sample size — written down (*pre-registration*)
2. **During:** no early stopping because the result looks good (or bad)
3. **After:** effect size + confidence interval + exact p — planned outcome first, everything else labeled exploratory
4. **Decide** with more than the p -value: size, uncertainty, costs, benefits

Assignment 2: analyzing experiments

- Your personal version of an **A/B-test data set**
- Hand-out: **today** · Deadline: Group A Sun, Nov 8 · Group B Wed, Nov 11
- Debrief at the start of the following session

Everything you need is in **Chapter 4**. Report effect sizes and confidence intervals — not just *p*-values.

Learning check

1. A checkout test: +2.40 €, 95% CI $[-0.30 \text{ €}; 5.10 \text{ €}]$, $p = 0.08$. Significant? One correct sentence for the head of e-commerce?
2. A new ad creative on 5 million users: CTR 0.410% $\rightarrow$ 0.415%, $p < 0.001$. Roll it out?
3. A colleague stops the test on the first day with $p < 0.05$. What is wrong?
4. Which test? (a) Conversion rates of three newsletter versions (b) Ratings (1–7) of the same 40 customers for two prototypes (c) Basket size, discount banner vs. none, 3,000 customers per group

Next session

Module 5 · Observational data I — *What if we cannot randomize?*

- Confounding, revisited — with data
- Regression as adjustment
- From correlation to (careful) causal claims